

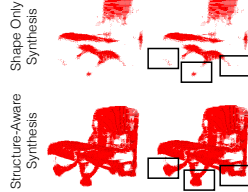
Structure-Aware Shape Synthesis

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Goal

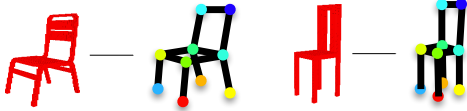
Synthesize realistic shapes that respect structure



Approach

Structure Decoder

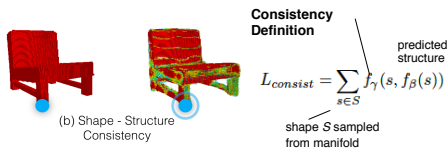
Given a shape, generate its structure



Shape Structure Consistency

Compute consistency between a shape and a structure

max. of shape voxel grid within a Gaussian sphere around the landmark coordinate

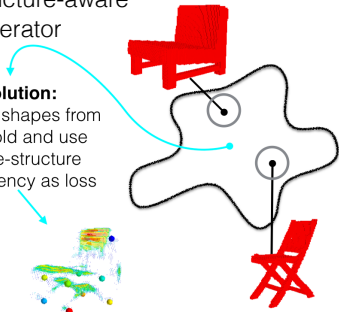


Consistency-Aware Training

Train a structure-aware shape generator

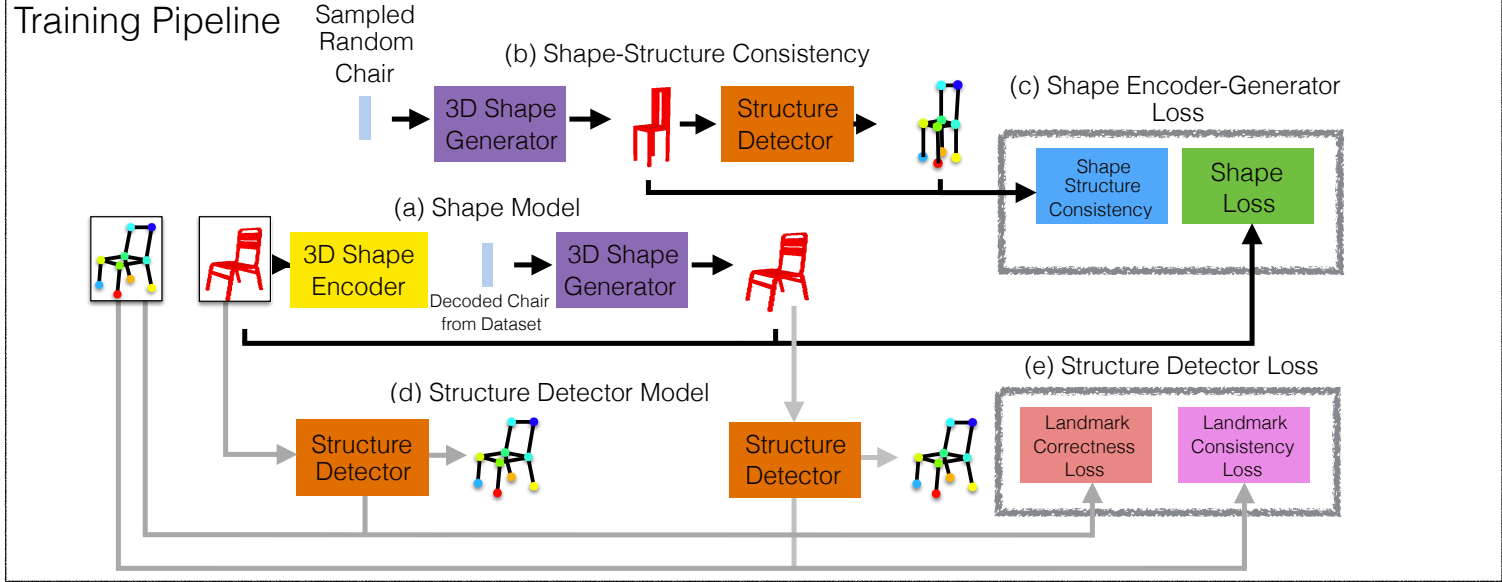
Solution:

Sample shapes from manifold and use shape-structure consistency as loss



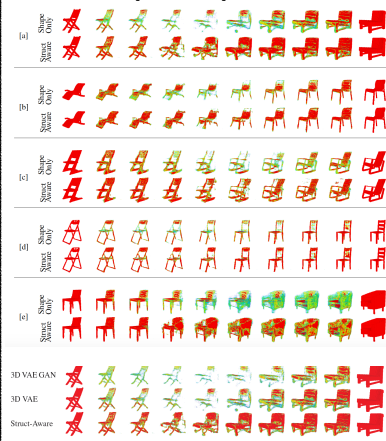
Evaluate shape-structure consistency loss $L_{consist}$

Training Pipeline

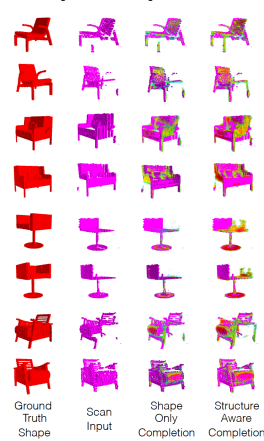


Applications

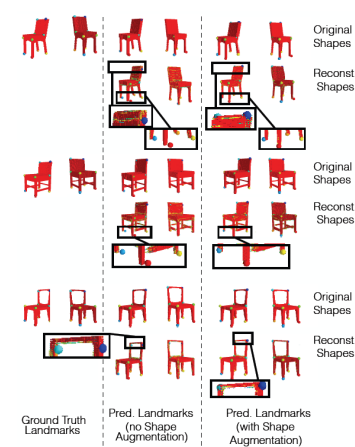
Shape Interpolation



Shape Completion



Landmark Prediction



Evaluation

Metric	Shape Only	Structure - Aware
Dense input	0.22	0.25
Sparse input	0.05	0.14

Table 1. Quantitative evaluation on the scan completion task. The reported numbers are average IoU computed with respect to the ground truth shape corresponding to the scan.

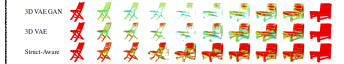


Figure 1. Comparison between 3D VAE-GAN [1] (an adversarial approach), 3D VAE, and our Structure-Aware method. The GAN model - generates slightly less noisy results - does not always generate structurally meaningful outputs.

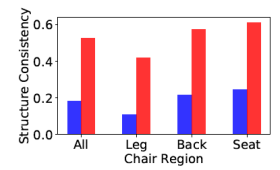


Figure 2. Evaluation of average shape structure consistency scores by chair region. The shape structure consistency score is in range [0, 1], where higher numbers indicate better performance. Note that for both methods, the most difficult region is the leg area, and the easiest region is the seat area.

Bibliography

- [1] J. Wu, T. Xue, J. J. Lim, Y. Tian, J. B. Tenenbaum, A. Torralba, and W. T. Freeman. Single image 3D interpreter network. In European Conference on Computer Vision, pages 365–382. Springer, 2016.
- [2] R. Girshik, D. F. Fouhey, M. Rodriguez, and A. Gupta. Learning a predictable and generative vector representation for objects. In European Conference on Computer Vision pages 484–499. Springer, 2016.
- [3] L. Ge, H. Liang, J. Yuan, and D. Thalmann. 3d convolutional neural networks for efficient and robust hand pose estimation from single depth images. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, volume 1, page 5, 2017.